CS401 Mega MCQs File Spring 2020 of Grand Quiz, Quiz 1, Quiz 2 BY MCS of Virtuallians

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Grand Quiz Spring 2020:

1. The base pointer accesses local variables using _____ offsets.

   Negative

2. Which of the following describes the purpose of MOVS instruction?

   Move memory to memory

3. Which part of this (0000000B80500) encoded instruction is an offset?

   0500

4. Stack is a data structure that behaves a first in last _____ manner.

   Out

5. In the instruction “mov word [es:160], 0x1230”, 30 represents _____ character.

   0

6. Multiplying two 4 bit numbers result in a _____ bit number.

   8

7. In case of near jump, the relative address is stored in _____ bits.

   16

8. _____ instructions have two parameters, one is the general purpose register to be loaded and the other is the memory location from which to load these registers.

   LDS

9. Physical memory address is of

   20 bits

10. _____ ports which interface the processor to the external world, including keyboards, mice, monitors, disc drives.

    Input, output

11. In base+offset addressing, the value contained in the base register is add with offset to get _____.

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Effective address

12. In 8051, there is an _____ stack.

Incrementing

13. AX register can be divided into _____ and _____ bytes

Lower, higher

14. CLI stands for

Clear the interrupt flag

15. When a 32 bit number is divided by a 16 bit number, the remainder is of

16 bits

16. MUL instruction performs an unsigned multiplication of _____ with the source operand.

Accumulator

17. DW can store _____ bit value in it.

16

18. When the stack pointer, points to the return address?

When the bubble sort subroutine is called

19. IAPX88 stands for _____.

Intel Advanced Processor Extensions 88

20. 90 is the op-code of

Do nothing

21. When characters are stored in any high level or low level language, the actual thing stored in a byte is their _____.

ASCII code

22. We can convert any digit to _____ by adding 0x30 in the digit.

ASCII

23. A complete _____ is called a pass over the array

Iteration

24. Which of the following is a non-destructive AND operation?

Test

25. In _____ operation the carry flag is inserted from the right causing every bit to move one location to its left and the most significant bit occupying the carry flag.

Rotate Through Carry Left (RCL)

26. ASCII table is the contiguous arrangement of the uppercase alphabets (41-5A), the lowercase alphabets (61-7A), and the numbers _____.

30-39

27. _____ can also be used as a masking operation to invert selective bits.

XOR

28. BH register is a _____ bit register.

8

29. Which of the following is the renamed version of conditional jump JZ?

JE

30. SP is associated (by default) with _____.

SS

31. The maximum amount of memory accessible using 8085 processor is _____.

64 KB

32. In XOR operation the output is 1 if

Both inputs are different

33. The clear screen operation initializes whole block of video memory to:

0720

34. The 8088 processor divides interrupts into _____ classes.

Two

35. Which of the following directive is used to reserve a 8 bit space in the memory for holding data?

db

36. All mathematical and logical operations are performed on the _____

Accumulator

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37. _____ jump is not position relative but is absolute

Far

38. Which of the following bit that "Shift Logical Right" operation copies in the carry flag?

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Right most bit

39. Which of the following register is used to hold address of the next instruction to be executed?

Program counter

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40. Group of bits processor uses to inform memory which element to read/write is collectively known as

Address bus

41. _____ containing the address of the next instruction to be executed.

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Instruction pointer (IP)

42. To convert the case of a character, we add or subtract _____ from its ASCII code.

0x20

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43. Which of the following instruction is effectively same as to multiply the value of AX by 8?

SHL AX, 8

MUL AX,3

44. _____ interrupts are those which occur side by side with some other activity.

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Synchronous

45. During CALL operation, the current value of the _____ is automatically saved on the stack, and the destination of CALL is loaded in the instruction pointer.

Instruction pointer

46. In SCAS Example, we use SCASB with _____ and a zero in AL register to find a zero byte in a string

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REPNE

47. In interrupt vector table. Introducing a new entry in this mapping table is called _____ an interrupt.
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Hooking

48. What does the following instruction do?
ADD AX. BX
Add both registers and load value into ax register

49. The process through which the segment register can be explicitly specified as known as
Segment addressing

50. REPE and REPNE prefixes are only meaningful with _____.
CMPS

51. _____ refers to the total number of bits in a memory cell.
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Cell width

52. The _____ operation is about shifting every bit one place to the right with a copy of the most significant bit left at the most significant place. The bit dropped from the right is caught in the carry basket.
Shift Arithmetic Right (SAR)

53. _____ and _____ cannot be used as 8bit register pairs like AX, BX, CX, and DX.
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SI, DI

54. AX and BX both are 16-bit register, if we perform AND operation on these two registers, then how many AND operations will be performed?
16 And operation

55. 8085 can access up to _____ of memory, whereas 8088 can access up to _____ of memory.
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64Kb, 1Mb

56. CS and IP are both _____ bit registers.
16

57. In 8080, there is a _____ stack.
Decrementing

58. An important role of the stack is in the creation of ____variables that are only needed while the subroutine is in execution and not afterward.
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Local

59. ____ movement of data is not allowed in assembly language.
Memory-to-Memory

60. With the execution of CALL instruction, the value of____ is decremented by 2.
SP

61. In interrupt vector table, introducing a new entry in the mapping table is called____ an interrupt.
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Hocking

62. Which of the following is the most illegal instruction?
    Mov al, ax
63. Motorola follows____
    Big endian
64. Which of the following instruction allows code reusability in 8088?
    CALL
65. When the first thing popped off from the stack, the stack would be the return "address" and not the ____
    Argument
66. ___ decrements SP (the stack pointer) by two and then transfers a word from source operand to the top of stack now pointed to by SP,
    PUSH
67. Which of the following is a Program Control Instruction?
    cmp ax,0
68. Logical addressing is a mechanism to access ___ memory.
    Physical memory
69. In assembly language "JN2" is used to
    Jump if the zero flag is not set
70. In segmented memory model, the size of one window is restricted to _____.
    64 KB
71. Twenty-bit register is formed by the combination of two ___ bit register.
    Sixteen
72. MOV[BX+SI+300],AX is a _____ addressing mode instruction.
    Base + index + offset
73. Physical address calculating depends on
    Effective address
74. There are ____ registers in iAPX88 architecture that can hold address of data.
    4
75. ____ also known as source operand since the data is moving to stack from this operand.
    PUSH
76. By default CS is associated with
    IP
77. The stack pointer contains the address of the word that is currently on _____.
    Top the stack
78. If AX=OOFF, then which of the following instruction can be used to change the value of AX to FFOO
    ANDAX, FFOO
79. All addressing mechanisms in iAPX88 return a number called ___ address.
    Effective

80. In 8088 processor, interrupts are divided into the following classes.
    Software Interrupts, Hardware Interrupts
81. Which of the following is the interrupt number for NMI?
    INT 3
82. There are ___ registers in IAPX88 architecture that can hold address of data.
    4
83. Use of AND operation to make selective bits zero in its destination operand is known as_____.
    Selective Bit Clearing
84. Standard ASCII has ____ characters?
    128
85. ____ is used to store both the instructions to be executes by the microprocessor and the data to be used in the computation.
    Microprocessor
86. Number of operands of ADC (add with carry) register are:
    3
87. DX play an important role in arithmetic Addition.
88. Stack is a Data Structure
89. REPE or REPNE are used with the SCAS instructions
90. LDS instruction have two parameters, one is the general purpose register to be loaded and other is the memory location from which to load these registers
91. Keywords used to define two bytes program DW
92. The shift logical left operation is the exact opposite of shift logical right
93. Sending the appropriate signal on the control bus to the memory is the responsibility of Control Bus
94. A parallel port has 2 views
95. The mechanism used to drop carry for making the calculated address valid is known is address wraparound
96. In Both SHL and SAL, a zero is inserted form right and every bit moves one position to its left wth most significant but dropping in to carry flag
97. The reduction in code size and the improvement in speed are the two reasons why block processing instruction were introduced in the 8080 Processor
98. Mov ax, [NUM1] is a 16 bit move instruction
99. Which of the following is the interrupt number for debug interrupt, INT 3
100. Each entry of the interrupt vector table is of 4 bytes
101. If BL contains 000000101 then after a Singe Right Shift, BL will contain 00000010
102. AND can be used to check weather particular bit of number are set or not

103. When the relative address stored with the instruction is in 16 bit , the jump is called a Near jump
104. The Stack of 8088 works on Word Sized element
105. The interrupt call loads new values in Flag segment
106. Mov AX, 0XB800, Move ES, AX : this instruction points ES to Video Base
107. When the operant of DIV instruction is of 16 bits then implied dividend will be of 8 bits
108. Which bit is attribute but representing the blue component of foreground color 0
109. When the operand of DIV instruction is of 16-bits then implied dividend will be stored in AX Register
110. Constant can never be used as destination
111. DB-25 is a Parallel Port Connector
112. Flag register is a special register in every architecture ,, is as also known as Program Status Word
113. BP stands for base pointer
114. Intel follows little endian
115. Mov [1234].ax is an example of **direct addressing**
116. **OR** is used to clear any specific bit in a binary number
117. In general the memory cell cannot be wider than the width of the data bus.
118. mov [bx], ax moves the two byte contents of the AX register to the address contained in the BX register in the current data segment
119. source operand always resided in accumulator register
120. INT instruction takes 1 byte argument varying from 0-255.
121. Program consists of 3 logical parts
122. 8088 provides a mechanism for mapping interrupts to interrupt handlers is called hooking an interrupt.
123. The routine that executes in response to an INT instruction is called the interrupt service routine (ISR) or the interrupt handler.
124. The push operation copies its operand on the stack , while the pop operation makes a copy from the top of the stack into its operand.
125. ROR : in the rotate right operation every bit moves one position to the right and the bit dropped from the right is inserted at the left and also copied into the carry flag.
126. The segment, offset pair is called a logical address
127. the **local variables** and the **parameters** are always stored in stack segment

128. SCAS compares a source byte or word in register AL or AX with the destination string element addressed by ES:DI and updates the flags.
129. JNP and JPO is taken if the last arithmetic operation produced a number in its destination that has
odd parity
130. JP and JPE is taken if the last arithmetic operation produced a number in its destination that has
even parity.
131. There are two forms of the DIV instruction.
132. Unconditional jump always transfer the control
133. The group of bits that the processor uses to inform the memory about which element to read or write is collectively known as the **address bus.**
134. ADC has three operands
135. In direct addressing the memory address given in the instruction is **fixed**
136. In which of the following addressing, the memory address is fixed and is given in the instruction? Direct
137. **DI and SI** pair of registers used to access memory
138. Total number of cells is called the **depth**
139. **Shift Logical Right (SHR)** copies the **right most bit** in the carry flag
140. REP with **MOVS** will utilize the full processor power to do the scrolling in minimum time.
141. The correlation process from the interrupt number to the interrupt handler uses a table called **interrupt vector table**
142. POP is also known as **destination operand**
143. The **parallel** port connector is a 25pin connector called **DB-25**
144. There are just **5** block processing instructions in 8088.
145. Interrupts are **asynchronous** and unpredictable
146. **CALL** instruction allows code reusability in 8088
147. Program Control Instructions
cmp ax, 0
148. In MULTIPLICATION ALGORITHM ,We take the first digit of the multiplier and multiply it with the **multiplicand**
149. Intel CPUs are **little-endian**
while **Motorola** 680x0 CPUs are **big-endian.**
150. **JO** jump is taken if the last arithmetic operation changed the sign unexpectedly.
151. **CLI** is a special instructions

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152. the interrupt call loads new values in CS
153. A 32bit processor has an accumulator of 32 bits.
154. Left shift on hexa-decimal number 9C40 ans is 0x13880
155. Each entry of the table is of four bytes
156. **VGA**. Video Graphics Adapter
157. The instruction “mov [bp], al” moves the one byte content of the AL register to the address contained in the BP register in the current **stack segment.**
158. STI stands for Set Interrupt Flag
159. Flags register is a special register in every architecture, it is also known as **program status word**
160. A special register exists in every processor called the **program counter or the instruction pointer**
161. mov word [es:160], 0x1230 12 meaning green color on blue background.
162. **DW** can store 16 bits
163. two variants of STOS are STOSB and STOSW
164. Another important role of the stack is in the creation of **local variables** that are only needed while the subroutine is in execution and not afterwards.
165. In Far jump both offset and segment are given
166. to multiply a number in register by 2 the number is shift left one bit
167. In case of downward compatible mechanism, the codes written for 8080 are Valid for 386 processor
168. Interrupt hooking is the mechanism that is used for mapping interrupt to interrupt handler
169. Which of the following statement is used to clear the value of AX register, xor ax, 0
170. Which assembly instruction is used to ADD data at address 1200 to bx register add bx, [1200]
171. Chose the correct option from the following addressing modes , from which both register moves into the data segment base+offset
172. RCR operation , a carry flag is inserted from left moving every bit one position to the right, with the right most bit is dropped in carry flag
173. What does the given instruction do?
SHR DL,1 Moves right most bit in carry
174. After the execution of SAR instruction most significant bit retains its original value
175. Control unit is the part of microprocessor that ménages the execution of instruction

176. In a comparison, if the both operands are same , the result of subtraction will be zero and the zero flag will be set
177. When SI and DI are used, we name the method Indexed Addressing
178. Which combination will you prefer to obtain the physical address of the stack SS:SP
179. Whenever we need access to a memory location whose address is not know until run time we use index register
180. Interrupts are asynchronous event
181. During program execution , if any change in AH or AL is reflected in AX as Well
182. Basic function of register is to hold operand
183. Which among the following is the pointer registers? index pointer and decession pointer
184. mov [bx], ax moves the two bytes content of AX register to the address contained in BX register in the current data segment
185. in string manipulation the instruction to clear the direction flag is CLD
186. If A is subtracted from B and the resulting answer is negative figure it means B is ? small number
187. in ___ operation the output is 1 only if only if both inputs are 1 ? AND
188. The interrupt call loads new values in CS, IP, and FLAGS.

Quiz 1 Spring 2020:

1. In programmable interrupt controller which of the following ports is referred as interrupt mask register?
    a. 19
    b. 20
    c. 21
    d. 22
2. _____is the highest priority interrupt in interrupt controller
    a. IRQ 0
    b. IRQ 1
    c. IRQ 2
    d. IRQ 3
3. IRET returns on the basis of _____ and _____
    a. CS, IP
    b. DS, IP
    c. CS, SS
    d. IP, SP

4. If _____ is set, the after every instruction a type 1 interrupt will be automatically generated.
    a. Parity flag
    b. Trap flag
    c. Carry flag
    d. Overflow flag
5. The interrupt handler uses _____instruction to return back to the caller.
    a. IRET
    b. RETI
    c. INTR
    d. RET
6. Which of the following interrupt is used for Arithmetic overflow?
    a. INT 1
    b. INT 2
    c. INT 3
    d. INT 4
7. During the execution of INT instruction , some contents are pushed on to the stack , the order of pushing then is
    a. CS , IP and then FLAGS register
    b. IP,CS and then FLAGS register
    c. FLAGS register, CS and then IP
    d. FLAGS register, IP and the CS
8. Each of the bits at port _____corresponding to one of the IRQ lines.
    a. 18
    b. 19
    c. 20
    d. 21
9. Which of the following pins of the parallel port connector are grounded?
    a. 10-18
    b. 18-25
    c. 25-32
    d. 32-39
10. Which of the following interrupts plays the most significant part during step debugging of a program?
    a. INT 0
    b. INT 1
    c. INT 2
    d. INT 3
11. Which of the following IRQs is connected to serial port COM 2?
    a. IRQ 0
    b. IRQ 1
    c. IRQ 2
    d. IRQ 3

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12. Which of the following IRQs is connected to serial port COM 1?
    a. IRQ 4
    b. IRQ 5
    c. IRQ 6
    d. IRQ 7
13. Which of the following IRQs is used by the parallel port?
    a. IRQ 4
    b. IRQ 5
    c. IRQ 6
    d. IRQ 7
14. In programmable interrupt controller which of the following ports is used to selectively enabling or disabling interrupts?
    a. 19
    b. 20
    c. 21
    d. 22
15. The space where all the registers of task are stored is called _____
    a. Control block
    b. Process control block
    c. Stack
    d. Memory
16. Which of the following is the destination register in IN instruction?
    a. AL or AX
    b. BL or BX
    c. CL or CX
    d. DL or DX
17. Which of the following interrupts is used for saving and restoring the registers
    a. INT 6
    b. INT 7
    c. INT 8
    d. INT 0
18. In multitasking environment, which of the following structure is used to maintain the ordering of active PCBs?
    a. Array
    b. Register
    c. Linked List
    d. Stack
19. In multitasking which of the following interrupts is used as scheduler during context switching?
    a. INT 21
    b. INT 16
    c. INT 13
    d. INT 8

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20. Which of the following are required for thread entry?
    a. CS and DS
    b. CS and IP
    c. IP and general purpose registers
    d. SS and SP
21. The time interval between two timer tick is?
    a. 40 ms
    b. 45 ms
    c. 50 ms
    d. 55 ms
22. Which of the following flag can be used for mathematical operations?
    a. Direction Flag
    b. Carry Flag
    c. Trap Flag
    d. Interrupt Flag
23. In multi tasking, the process of saving and restoring to values of registers from a process control block (PCB) is called _____
    a. Context switching
    b. Context saving
    c. Context restoring
    d. Code switching
24. When two devices in a system want to use the same IRQ line, is referred as:
    a. IRQ Collision
    b. IRQ Conflict
    c. IRQ Crash
    d. IRQ Blockage
25. An End of Interrupt (EOI) signal is sent by
    a. Handler
    b. Processor
    c. IRQ
    d. PIC
26. Which of the following is the BIOS interrupt providing keyboard service?
    a. INT 0x13
    b. INT 0x14
    c. INT 0x15
    d. INT 0x16
27. Threads can have function calls, parameters and _____ variables.
    a. Global
    b. Local
    c. Legal
    d. Illegal
28. Which of the following arranges jobs in a sequence in order to be executed?
    a. Process control block

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b. Arranger
c. Control unit
d. Scheduler

29. Which of the following flags cannot be cleared using an assembly instruction?
a. Trap flag
b. Interrupt flag
c. Direction flag
d. Carry flag

30. Which of the following is the highest priority interrupt?
a. INT3
b. INT2
c. INT1
d. INT0

31. At the end of servicing an interrupt, which of the following is used to inform tha PIC that it is completed?
a. RET
b. EOI
c. IRET
d. RET IN

32. Which of the following interrupts is used for maintaining the system time?
a. INT 0
b. INT 1
c. INT 8
d. INT 10

33. Which of the following is used for exporting parallel port services?
a. INT 17
b. INT 16
c. INT 15
d. INT 8

34. In programmable interrupt controller which of the following ports is referred as a control port?
a. 19
b. 20
c. 21
d. 22

35. The first parallel port LPT1 has port number designated from _____.
a. 178 to 17A
b. 278 to 27A
c. 378 to 37A
d. 478 to 47A

36. _____ is/are the port number(s) for parallel port.
a. 20 and 21
b. 60 to 64

c. 380
d. 378

37. The number of pins in a parallel port connector are?
a. 20
b. 25
c. 30
d. 35

38. The offset address of an interrupt n will be at
a. n
b. nx2
c. nx3
d. nx4

39. Programmable interrupt controller has two ports (20 and 21). Port 20 is the control port while port 21 is _____.
a. The interrupt mask register
b. Interrupt port
c. Output port
d. Input port

40. Which of the following instruction selects memory address space?
a. MOV
b. DEC
c. IN
d. ADD

41. The thread registration code initializes the PCB and adds it to the linked list so that the _____ will give it a turn.
a. Assembler
b. Scheduler
c. Linker
d. Debugger

42. Which of the following instruction is used for disabling all the interrupts during a program execution?
a. Cli
b. Sti
c. Reti
d. Iret

43. IBM AT has how many PICs (Programmable interrupt controller)
a. 1
b. 2
c. 3
d. 4

44. Programmable interrupt controller (PIC) has
a. One input signals and eight output signals
b. One input signal and one output signal

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c. Eight input signals and one output signals
d. Eight input signals and eight output signals

45. In 8088 processor, there can be total _____ possible entries in an interrupt vector table.
a. 256
b. 64
c. 128
d. 512

46. Each thread can have their own
a. Execution area
b. Stack
c. Memory
d. Array

47. The parallel port connector is called?
a. DB-25
b. BD-25
c. DB-24
d. BD-24

48. In 8088, the physical memory address for Interrupt Vector Table is fixed and the table occupies exactly _____ of memory.
a. 1 KB
b. 1 MB
c. 64 KB
d. 4 Bytes

49. Which of the following interrupts is non maskable interrupt?
a. INT 0
b. INT 1
c. INT 2
d. INT 3

50. Using OUT instruction on parallel data port results into a signal of _____ for every 1 bit.
a. 0 V
b. 1 V
c. 5 V
d. 10 V

51. PCB stands for?
a. Process Control Block
b. Process Cleaning Block
c. Programmable Counter Block
d. Programs Control Block

52. Which of the following interrupts is used in debugging with the trap flag?
a. INT 0
b. INT 1

c. INT 2
d. INT 3

53. Which of the following port number is used to send an end of interrupt (EOI) signal to the PIC after an interrupt is ended?
a. 0x16
b. 0x20
c. 0x60
d. 0x378

54. Which of the following instruction is used for reading a char from keyboard?
a. out al, 0x60
b. in al, 0x60
c. out dx, al
d. out dx, 0x378

55. In PIC, which of the following port is used for selectively enabling or disabling interrupts?
a. 19
b. 20
c. 21
d. 22

56. DOS has single entry point to access all its services through interrupt_____
a. INT 21
b. INT 16
c. INT 2
d. INT 10

57. Which of the following pins of parallel port are used to carry the data from processor to the device connected through parallel port?
a. Pin 2 to pin 9
b. Pin 10
c. Pin 4
d. Pin 18 to pin 25

58. The input frequency of the programmable interval timer(PIT) is
a. Fixed
b. Depends on processor block
c. Variable
d. Depends on hardware attached

59. All registers of task are stored in space called _____
a. Process storage block
b. Process control block
c. Process terminal
d. Swapping space

60. Which of the following IRQ`s is derived by a key board
a. IRQ 0
b. IRQ 1

c. IRQ 2
d. IRQ 3

61. The programmable interval timer (PIT) has input frequency of
a. 1.193MHz
b. 2.193MHz
c. 3.193MHz
d. 4.193MHz

62. Changing and restoring the state of Control processing unit (CPU) is referred as
a. Thrashing
b. Multitasking
c. Context Switching
d. Multithreading

63. For the external hardware to generate an interrupt there are how many pins outside the processor?
a. 1
b. 2
c. 3
d. 4

64. Which of the following instruction selects peripheral address space?
a. MOV
b. DEC
c. IN
d. ADD

65. Which of the following is the most commonly used port with printer?
a. Serial Port
b. USB port
c. Parallel Port
d. DVI Port

66. _____allows us to execute one instruction at a time rather than the whole program at once
a. INT 0
b. INT 1
c. INT 2
d. INT 3

67. Which of the following is used for exporting keyboard services?
a. INT 16
b. INT 11
c. INT 12
d. INT 8

68. TSR stands for
a. Terminate and store routines
b. Transmits and save resources
c. Terminate and Stay resident

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d. Truncate and start recursively

69. Which of the following interrupts is of highest priority interrupt?
    a. Key Board Interrupt
    b. Timer Interrupt
    c. INT 2
    d. INT 3
70. In older computers, the port number designed for parallel pert LPT2 are_____
    a. 078-07A
    b. 178-17A
    c. 278-27A
    d. 378-37A
71. Which of the following is the ACK pin in DB-25 Connector
    a. 10
    b. 11
    c. 12
    d. 13
72. Which of the following IRQs is a cascading interrupt?
    a. IRQ 0
    b. IRQ 1
    c. IRQ 2
    d. IRQ 3
73. In interrupt masking, we use _____ to get control from the program without letting the program know about it.
    a. IRQ 0
    b. IRQ 1
    c. IRQ 5
    d. IRQ 7
74. The output of programmable timer interval is connected to which line of the PIC?
    a. IRQ 0
    b. IRQ 1
    c. IRQ 2
    d. IRQ 3
75. Which of the following IRQs is used for Floppy disk drive?
    a. IRQ 4
    b. IRQ 5
    c. IRQ 6
    d. IRQ 7
76. In multitasking, each PCB requires _____ bytes to store the values of registers.
    a. 14
    b. 28
    c. 32
    d. 16
77. Which of the following IRQs is used for sound card or network card?

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a. IRQ 4
b. IRQ 5
c. IRQ 6
d. IRQ 7

78. Programmable interrupt controller has _____ ports.
a. 1
b. 2
c. 3
d. 4

79. Which of the following bits of parallel control port enables the IRQ 7 triggering by the ACK pin?
a. Bit 1
b. Bit 2
c. Bit 3
d. Bit 4

80. Which of the following IRQs is derived by a timer device?
a. IRQ 0
b. IRQ 1
c. IRQ 2
d. IRQ 3

81. _____ is a special type of interrupt that returns to the same instruction instead of the next instruction.
a. Divide overflow
b. Debug interrupt
c. Arithmetic overflow interrupt
d. Change of sign interrupt

82. All the registers & stack are saved in
a. Multitasking
b. Multi-Processing
c. Function call
d. BIOS

Quiz 2 Spring 2020:

1. In the context of video service, if we want to set the cursor at top left corner, which of the values will be stored in DH and DL respectively?
a. 00.00 page 141
b. 00.80
c. 25.80
d. 0.01

2. Which of the following INT 21 service is used to read character from standard input?

a. 00
b. 01 page 144
c. 02
d. 03

3. "INT13-BIOS disk service" generally uses which register to return the "error flag"?
   a. CF page 148
   b. DL
   c. AH
   d. AL
4. Which of the following service of INT 10 is used to get the video font information?
   a. 1129h
   b. 1130h page 142
   c. 1128h
   d. 1127h
5. Which of the following service of INT 10 is used to write graphic pixel on the screen?
   a. 0x0A
   b. 0x0B
   c. 0x0C page 144
   d. 0x0D
6. On executing INT 0x21-service 0x3D, if file is successfully opened then
   a. CF will contain 1
   b. CF will contain 0
   c. ZF will contain 1
   d. ZF will contain 0
7. On executing INT 0x21-service 0x3D, if file can't be opened then
   a. CF will contain 1
   b. CF will contain 0
   c. ZF will contain 1
   d. ZF will contain 0
8. Device drivers can be divided into _____ major categories
   a. 5
   b. 4
   c. 3
   d. 2

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9. We can set the current file position in DOS using service number _____
    a. 0x38
    b. 0x40
    c. 0x42 page 154
    d. 0x43
10. COM1 is accessible via ports _____ while COM2 is accessible via _____
    a. 3FD, 3F9
    b. 2F8-2FF, 3F8-3FF
    c. 3F9, 3FD
    d. 3F8-3FF, 2F8-2FF page 164
11. In the context of vides services, if we want to write string then which of the value will be placed in AH?
    a. 10h
    b. 12h
    c. 11h
    d. 13h page 142
12. In the context of video service, if we want to scroll the window up which of the following will be value of AH?
    a. AH = 05h
    b. AH = 06h page 141
    c. AH = 07h
    d. AH = 08h
13. In the context of video services, if we want to set cursor position which of the following will be the value of AH?
    a. AH = 01h
    b. AH = 02h page 139
    c. AH = 03h
    d. AH = 04h
14. Which standard format is used for sending data on the serial port?
    a. Full duplex
    b. BIOS INT 14
    c. RS232 page 163
    d. COM1
15. INT 0x13 service 0x08 is use to _____
    a. Read disk sector
    b. Write disk sector
    c. Reset disk sector

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d. Get drive parameter page 148

16. Parameters pushed to the routine in _____ order with the rightmost being pushed first using language
    a. Reverse page 179
    b. Proper
    c. Forward
    d. Same

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17. Parameters pushed to the routine in _____ order with the left most being pushed first using Pascal language
    a. Proper page 179
    b. Same
    c. Reverse
    d. Forward

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18. _____ symbol is added to any function or variable using the C language naming convention
    a. Underscore page 177
    b. #
    c. &
    d. @

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19. Which of the following BIOS INT provides serial port services
    a. INT 21
    b. INT 14 page 163
    c. INT 10
    d. INT 08

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20. A 32-bit address register can access up to _____ of memory
    a. 2 MB
    b. 2 GB
    c. 4 MB
    d. 4 GB page 167

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21. BIOS video services are exported via interrupt
    a. INT 10 page 139
    b. INT 9
    c. INT 8
    d. INT 7
22. Which of the following is used to terminate a string in assembly language?
    a. !
    b. #

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c. $
d. ;

23. The bits are sent one by one in specially formatted fashion on the_____
    a. Half duplex
    b. Parallel port
    c. Full duplex
    d. Serial port page 163

24. In VESA defined modes _____bits for every pixel are organized.
    a. 8 color
    b. 16 color page 172
    c. 32 color
    d. 64 color

25. What is the abbreviation VESA _____ and VBE _____
    a. Video Electrical System Association, VGA BIOS Extensions
    b. Video Electronics Standards Administration , Video Bus Extensions
    c. Video Electronics Standard Association, Video BIOS Extensions page 172
    d. VGA Electronics Standard Association, Video BIOS External

26. Which of the following services of INT 21 is used to write a string on standard output?
    a. 00h
    b. 02h
    c. 09h page 144
    d. 10h

27. In 9pin DB 9 connector, which pin is assigned to RD (Received Data)?
    a. 1
    b. 2 page 163
    c. 3
    d. 4

28. The maximum length of DOS command line parameters is
    a. 64 bit
    b. 127 bits
    c. 256 bits
    d. 512 bits

29. In C _____ is responsible for removing the parameters.
    a. Clear
    b. Callee

c. Caller page 179
d. Stack

30. While using INT 10 for writing graphic pixel on screen, which of the following registers hold the information about pixel color?
    a. DH
    b. DL
    c. AH
    d. AL page 144

31. As compared to _____ , _____ provides more cooked services.
    a. BIOS, DOS page 144
    b. DOS, OS
    c. DOS, BIOS
    d. OS, DOS

32. _____ Register is used for storing the base and limit of GDT.
    a. LGDT
    b. GDTR page 168
    c. Null Descriptor
    d. LDT

33. INT13-BIOS disk service" generally uses which register to return the "error code"?
    a. CF
    b. DL
    c. AH page 148
    d. AL

34. In case of COM File, first command line parameter is stored at _____ offset of 'Program segment prefix'
    a. 0x80
    b. 0x82
    c. 0x84
    d. 0x86

35. New devices are allowed to work with the existing operating system with the help of _____.
    a. Sockets
    b. Port addresses
    c. BIOS
    d. Device drivers page 158

36. _____ byte of DOS data buffer stores the maximum characters buffer can hold.
    a. First page 144
    b. Second
    c. Third
    d. Last
37. The register size of 386 architecture is _____
    a. 16 bit
    b. 32 bit page 167
    c. 20 bit
    d. 4 GB
38. Com 1 is connected with
    a. IRQ 2
    b. IRQ 3
    c. IRQ 4 page 106
    d. IRQ 5
39. Which of the following services of INT 10 is used for writing the string?
    a. 0x12
    b. 0x13 page 142
    c. 0x11
    d. 0x10
40. The first entry of the GDT must always be zero, it is called the _____
    a. Null Descriptor page 168
    b. Data Descriptor
    c. Code Descriptor
    d. Global Descriptor Table
41. The first sector on hard disk contains the
    a. Hard disk
    b. Partition table page 150
    c. Data size
    d. Sector size
42. BIOS video services are broadly classified into _____categories
    a. 1
    b. 2 page 140
    c. 3
    d. 4

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43. In case of COM file, maximum length of parameters passed through command line can be _____
    a. 63 bytes
    b. 127 bytes
    c. 255 bytes
    d. 511 bytes
44. In Pascal _____ is responsibility for removing the parameters
    a. Stack
    b. Clear
    c. Caller
    d. Callee page 179
45. Hard disk MBR (Master Boot record) is of size
    a. 446 bytes page 150
    b. 350 bytes
    c. 512 bytes
    d. 256 bytes
46. In 9pin DB 9 connector , which pin is assigned to Signal Ground
    a. 3
    b. 4
    c. 5 page 163
    d. 6
47. Two prevalent calling conventions are the _____ and _____
    a. C# calling convention, Java calling convention
    b. C calling convention, Pascal Calling convention page 179
    c. C++ calling convention, Java calling convention
    d. C++ calling Convention, Python calling Convention
48. Com2 is connected with
    a. IRQ 2
    b. IRQ 3 page 106
    c. IRQ 4
    d. IRQ 5
49. BPB stands for
    a. Basic Parameter Block
    b. BIOS Precise Block
    c. Basic Precise Block
    d. Bios Parameter Block page 159

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50. While writing graphic pixel on the screen, which registers hold the value for setting cursor position?
    a. AX, BX
    b. AX, CX
    c. BX, CX
    d. CX, DX

51. The VESA VBE 2.0 standard include a _____ mode which allows direct access to the whole video memory.
    a. Parallel frame buffer
    b. Liner frame buffer page 172
    c. Real
    d. Protected

52. In 9pin DB 9 connector, which pin is assignment to TD (Transmitted Data)?
    a. 1
    b. 2
    c. 3
    d. 4

53. INT 0x13 service 0x03 is used to _____
    a. Read disk sector
    b. Write disk sector page 148
    c. Reset disk sector
    d. Get drive parameter

54. Serial port services are provided by the _____
    a. BIOS INT 11
    b. BIOS INT 14 page 163
    c. BIOS INT 13
    d. RS232

55. Which of the following register hold the page number for using the write string service of INT 10?
    a. AH
    b. BL
    c. CL
    d. BH page 144

56. The “program segment prefix” for com file is of _____size
    a. 64bytes
    b. 128 bytes
    c. 256 bytes

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d. 512 bytes

57. Which of the following registers are used as DOS input buffer
    a. DS, AX
    b. SS, SP
    c. DS, DX page 144
    d. CS:BX

58. Which register is generally used to specify the service number of an interrupt
    a. DX
    b. AX page 100
    c. BX
    d. CX

59. The root directory of floppy contains _____fixed entries
    a. 64
    b. 256
    c. 128
    d. 512 page 156

60. In FAT32 file system directory entry in DOS consist of how many bytes?
    a. 16
    b. 24
    c. 32 page 147
    d. 64

61. _____ is the highest priority interrupt in interrupt controller
    a. IRQ 0 page 106
    b. IRQ 1
    c. IRQ 2
    d. IRQ 3

62. Operating system organize data in form of
    a. Folder
    b. Batch file
    c. File page 153
    d. None of given

63. The higher 16 bit of EAX register are labeled as
    a. AX
    b. EAH
    c. AH
    d. None of given

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